Canvas Quiz 2, Modules 2–3 (graded)

Module 2 - IV Bolus Administration · Canvas quiz · No date

Name _____ Date _____

Question 1 Which of the following best describes the volume of distribution?

A. hypothetical volume of body fluid that would be required to dissolve the total amount of drug at the same concentration as that found in the blood
B. proportionality constant relating the rate of drug absorption to the rate of drug elimination
C. estimate of the time course of drug absorption, distribution, metabolism, and elimination
D. physiological measure of the volume occupied by the peripheral compartments

Question 2 A 250-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug. A = 10.16 mg/L, B = 5.65 mg/L, $\alpha = 3.59\ hr^{-1}$, $\beta = 0.15\ hr^{-1}$. What is the elimination half-life of this agent? Round to the nearest hundredth (two decimal places).

Question 3 A 250-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug. A = 10.68 mg/L, B = 3.13 mg/L, $\alpha = 3.55\ hr^{-1}$, $\beta = 0.15\ hr^{-1}$. What is the concentration of the drug in the plasma 4 hours following administration of the dose? Round to the nearest hundredth (two decimal places).

Question 4 Why is a loading dose used?

A. To immediately obtain a therapeutic plasma concentration
B. To reduce the elimination half-life
C. To reduce the adverse effects associated with the drug
D. To improve patient adherence

Question 5 An antibacterial agent with a volume of distribution of 22 L and a clearance of 3.2 L/hr was infused intravenously until a concentration of 14 mg/L was reached. What was the concentration of drug in the plasma 5 hours following cessation of the infusion? Round to the nearest tenth (one decimal place).

Question 6 Which of the following results when the rate of infusion is increased?

A. Steady-state concentration decreases
B. The time to reach steady-state decreases
C. The time to reach steady-state increases
D. Steady-state concentration increases

Question 7 Recommend an infusion rate to achieve a steady-state concentration of 23 mcg/mL for an agent with an apparent volume of distribution of 15 L and half-life of 4 hr. Round to the nearest whole number.

Question 8 An analgesic drug is to be administered by continuous intravenous infusion at a rate of 38 mg/hr to a 60 year-old, 154 pound female patient. The drug has a half-life of 4 hours and an apparent volume of distribution of 0.21 L/kg. What is the expected steady-state concentration? Round to the nearest tenth (one decimal place).

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: A. hypothetical volume of body fluid that would be required to dissolve the total amount of drug at the same concentration as that found in the blood

This is the definition on her slide, word for word. The volume is hypothetical because it is the dose divided by the plasma concentration, not a space that can be measured, which is why it can exceed total body water.

Question 2 Worked by the drill

Answer: 4.62 hr

WHAT THE STEM GIVES

$\mathbf{D_0}$	250-mg IV bolus	*not needed: half-life does not depend on the dose*
n	six healthy volunteers	*not needed*
A	10.16 mg/L	*not needed: intercepts do not set the half-life*
B	5.65 mg/L	*not needed*
α	3.59 hr^{-1}	*not needed: the distribution phase is not the elimination half-life*
β	0.15 hr^{-1}	the denominator of $\frac{0.693}{\beta}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{\beta}$$

The elimination half-life of a two-compartment drug belongs to the terminal phase, whose slope is β. The distribution constant α describes the early fall that ends once the tissues have filled, and A and B are intercepts, not rates, so none of them enters.

ALGEBRA

$$t½ = \frac{0.693}{0.15\ hr^{-1}} = 4.62\ hr$$

Dividing the pure number 0.693 by β in reciprocal hours leaves hours, because hr^{-1} in the denominator inverts to hr: $\frac{0.693}{0.15} = 4.62$. The dose of 250 mg and the intercepts A and B are not used; they would enter only for a volume or an initial concentration.

ROUND

4.62 hr

The quotient 0.693 divided by 0.15 is exactly 4.62, so rounding to the nearest hundredth changes nothing. Her key is 4.62 hr with a margin of 2%, which is 0.09 hr either way, so any answer from 4.53 to 4.71 scores. Two decimal places were asked for.

DOES THE NUMBER MAKE SENSE?

β = 0.15 hr^{-1} removes about 15% an hour, so halving takes between 4 and 5 hours: $\frac{0.693}{0.15} = 4.62$ hr. The elimination half-life uses the smaller exponent, β.

Question 3 Worked by the drill

Answer: 1.72 mg/L

WHAT THE STEM GIVES

D_0	250-mg IV bolus	*not needed: the equation already carries the dose*
A	10.68 mg/L	the coefficient of $e^{-\alpha t}$
B	3.13 mg/L	the coefficient of $e^{-\beta t}$
α	3.55 hr^{-1}	the exponent $\alpha t = 3.55 \times 4 = 14.2$
β	0.15 hr^{-1}	the exponent $\beta t = 0.15 \times 4 = 0.6$
t	4 hours	t in both exponentials

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = Ae^{-\alpha t} + Be^{-\beta t}$$

A two-compartment concentration is the sum of two exponential terms, one for each phase. Each intercept pairs with its own rate constant: A with α, B with β.

ALGEBRA

$$10.68 \times e^{-3.55 \times 4} = 10.68 \times e^{-14.2} = 0.0000$$

The distribution term has decayed to nothing by 4 hours, because α is large. Dropping it changes nothing at two decimal places, but writing it shows that it was considered rather than forgotten.

ALGEBRA

$$3.13 \times e^{-0.15 \times 4} = 3.13 \times e^{-0.6} = 3.13 \times 0.5488 = 1.7178 \text{ mg/L}$$

The terminal term carries the whole answer at 4 hours. The exponent is dimensionless because $hr^{-1} \times hr$ cancels.

ROUND

1.72 mg/L

Her key is 1.72 with a margin of 3%, so 1.67 to 1.77 scores. Rounding the exponent or $e^{-0.6}$ early stays inside that margin; rounding B does not matter since it is given.

DOES THE NUMBER MAKE SENSE?

By 4 hours the fast α term has vanished, so the level is the B term alone; 0.6 in the exponent leaves 0.5488 of the 3.13 mg/L B intercept: 1.72 mg/L, below 3.13 and above zero.

Question 4 Worked by the drill

Answer: A. To immediately obtain a therapeutic plasma concentration

An infusion alone takes three to five half-lives to reach steady state. A loading dose puts the steady-state amount, $C_{ss} \times V_D$ (the steady-state concentration times the apparent volume of

distribution), into the body at once, so the therapeutic level is there from the start and the infusion then holds it.

Question 5 Worked by the drill

Answer: 6.8 mg/L

WHAT THE STEM GIVES

V_D	22 L	the denominator of $k = \frac{Cl}{V_D}$
Cl	3.2 L/hr	the numerator of $k = \frac{3.2}{22} = 0.1455\ hr^{-1}$
C(stop)	14 mg/L	the level the decay starts from
t	5 hours following cessation	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

After the infusion stops, $C = C(stop) \times e^{-kt}$, with $k = \frac{Cl}{V_D}$

Once input ends, the drug falls by first-order elimination from whatever concentration it had reached, exactly like an IV bolus starting at that concentration. The stem gives no k, so it has to be assembled from clearance and volume.

ALGEBRA

$$k = \frac{3.2\ L/hr}{22\ L} = 0.1455\ hr^{-1}$$

Litres cancel, leaving reciprocal hours. The model is not named in the stem; one compartment is assumed as in every infusion problem of this module.

ALGEBRA

$$C = 14\ mg/L \times e^{-0.1455 \times 5} = 14 \times 0.4832 = 6.77\ mg/L$$

The exponent is dimensionless, and the 14 mg/L is the concentration at the moment the infusion was stopped, whether or not that was steady state.

ROUND

6.8 mg/L

Her key is 6.8 with a margin of 2%, so 6.66 to 6.94 scores. The unrounded 6.77 rounds to 6.8 at one decimal place.

> **DOES THE NUMBER MAKE SENSE?**
> $k = 0.1455\ hr^{-1}$ is a half-life a little under 5 hours, so 5 hours is just over one half-life and a bit under half of 14 mg/L remains: 6.8 mg/L, below 14.

Question 6 Worked by the drill

Answer: D. Steady-state concentration increases

$C_{ss} = \frac{R}{Cl}$, where C_{ss} is the steady-state concentration, R the infusion rate and Cl clearance. At the plateau the rate in, R, equals the rate out, $Cl \times C_{ss}$; clearance is a constant for the patient, so a

larger R must be balanced by a larger C_{ss}. Doubling the rate doubles C_{ss}.

Question 7 Worked by the drill

Answer: 60 mg/hr

WHAT THE STEM GIVES

C_{ss}	23 mcg/mL	written 23 mg/L, multiplied by clearance
V_D	15 L	in $Cl = kV_D = 0.1733 \times 15 = 2.599$ L/hr
t½	4 hr	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = C_{ss} \times Cl = C_{ss} \times k \times V_D$$

At steady state the rate in, R, equals the rate out, clearance Cl times the steady-state concentration C_{ss}, so $R = C_{ss} \times Cl$. Clearance is not given, so it is built as $Cl = k \times V_D$, where k is the elimination rate constant and V_D the apparent volume of distribution.

UNIT

$$23\ mcg/mL = 23\ mg/L$$

Multiplying numerator and denominator by 1000 turns mcg/mL into mg/L, so the concentration multiplies a volume in litres to give milligrams, which is what mg/hr needs.

ALGEBRA

$$k = \frac{0.693}{4\ hr} = 0.1733\ hr^{-1};\ Cl = 0.1733 \times 15\ L = 2.599\ L/hr$$

The first-order relation $t½ = \frac{0.693}{k}$ is rearranged to $k = \frac{0.693}{t½}$, so 0.693 divided by 4 hr gives 0.1733 hr^{-1}, the fraction of the drug removed each hour. Clearance is then $Cl = k \times V_D$: 0.1733 hr^{-1} times 15 L, and hr^{-1} times L gives L/hr, so 2.599 L/hr.

ALGEBRA

$$R = 23\ mg/L \times 2.599\ L/hr = 59.8\ mg/hr$$

The target concentration in mg/L multiplies the clearance in L/hr: 23 × 2.599 = 59.8. The litres in the denominator of mg/L cancel the litres in the numerator of L/hr, leaving mg/hr, an amount of drug per hour, which is what an infusion rate is.

ROUND

$$60\ mg/hr$$

Rounding 59.8 mg/hr to the nearest whole number gives 60 mg/hr, because 0.8 rounds up. Her key is 60 with a margin of 2%, which is 1.2 mg/hr either way, so any answer from 58.8 to 61.2 scores. A whole number was asked for.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance is 2.599 L/hr, and holding 23 mg in each litre cleared needs 23 × 2.599 = 59.8 mg/hr, so about 60 mg/hr. The rate is above zero.

Question 8 Worked by the drill

Answer: 14.9 mg/L

WHAT THE STEM GIVES

R	38 mg/hr	the numerator
age	60 year-old	*not needed: age does not enter C_{ss}*
weight	154 pound	converted: $\frac{154}{2.2}$ = 70 kg, scales V_D
t½	4 hours	gives k = $\frac{0.693}{4}$ = 0.1733 hr^{-1}
V_D	0.21 L/kg	scaled to 70 kg: 14.7 L; Cl = kV_D = 2.547 L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{Cl} = \frac{R}{k \times V_D}$$

At steady state the infusion rate R equals the elimination rate, which is clearance Cl times the steady-state concentration C_{ss}, so $C_{ss} = \frac{R}{Cl}$. Clearance is not given, so it is built as Cl = k × V_D, where k is the elimination rate constant and V_D the apparent volume of distribution.

UNIT

$$154 \frac{lb}{2.2} = 70 \text{ kg}; \quad V_D = 0.21 \text{ L/kg} \times 70 \text{ kg} = 14.7 \text{ L}$$

The volume is given per kilogram and the weight in pounds, so the weight is converted first and then scaled. Age and sex are stated and not used.

ALGEBRA

$$k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}; \quad Cl = 0.1733 \times 14.7 \text{ L} = 2.547 \text{ L/hr}$$

The first-order relation $t½ = \frac{0.693}{k}$ rearranges to $k = \frac{0.693}{t½}$: 0.693 divided by 4 hr gives 0.1733 hr^{-1}, the fraction removed each hour. Clearance is then k × V_D with the 14.7 L from the unit step: 0.1733 hr^{-1} × 14.7 L = 2.547 L/hr, since hr^{-1} times L is L/hr.

ALGEBRA

$$C_{ss} = \frac{38 \text{ mg/hr}}{2.547 \text{ L/hr}} = 14.92 \text{ mg/L}$$

The infusion rate in mg/hr is divided by the clearance in L/hr: 38 divided by 2.547 gives 14.92. The per-hour in the numerator cancels the per-hour in the denominator, leaving milligrams per litre, a concentration, which is what C_{ss} must be.

ROUND

14.9 mg/L

Rounding 14.92 mg/L to one decimal place gives 14.9 mg/L, because the second decimal, 2, rounds down. Her key is 14.9 with a margin of 2%, which is 0.3 mg/L either way, so any answer from 14.6 to 15.2 scores. One decimal place was asked for.

DOES THE NUMBER MAKE SENSE?

Each hour 38 mg enters and 2.547 L is cleared, so the plateau is $\frac{38}{2.547}$, about 15 mg/L: 14.9 mg/L, above zero.
